

第一届“认证杯”数学中国

数学建模国际赛

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Say No To Coffee Spilling

Abstract

In the fast-paced society, almost all of us choose to walk with a cup of coffee. While we often spill the coffee, this phenomenon is provided with a better result under the research of three types of cup in this paper.

We decompose this problem into two parts: human walking and liquid sloshing. For the velocity of walking changing with the time, we simplify the process into the synthesis of horizontal and vertical simple harmonic motion and test the correction of the model. For the liquid sloshing, we also simplify the changing of elevation angle into a simple harmonic motion and express this process through the liquid balance equation and mechanical equilibrium equation. We find that whether the coffee will spill or not is related to stride length, stride frequency, shape of cup and volume of coffee. What's more, we test the model by analysis of parameters. Through the fitting of parameters by least square method, to the focused man we calculate that the maximal deviation angle is 10.8 degree and the maximal capacity is 94.8% under the circumstance that the stride length is 0.65m, the period of step is 1.1s, and the average velocity is 0.85m/s.

In terms of robot, we add the random vibration of swing of hands into the previous model. The vibration has positive correlation with stride length and negative correlation with stride frequency. Through testing the assemble of parameters in the walking strategy, we calculate the velocity, maximal deviation of liquid interface and the maximal capacity of coffee under a suitable walking strategy in two types of cup. We find that the slender one matches the higher stride frequency while the dumpy one matches the conservative strategy. According to the strategy, we can guarantee the maximal velocity and capacity of coffee without spilling.

Key words: simple harmonic motion, least square method, liquid equilibrium equation

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1. Introduction

Under the busy and stressful circumstance, it's common to see white-collars walk with a cup of coffee. However, spilling the coffee and making a mess is inconvenient to us. The phenomenon causes the thinking on how to prevent coffee from spilling out the cup. Although the problem is too insignificant to mention and too hard to find the breakthrough point, it's useful for the robot at some degree.

Obviously, the problem is composed of two parts: human walking and liquid sloshing. In general, the factors which lead to the spilling coffee are probably related to stride frequency, velocity and compound factors.

On the basis of the biomechanics [1], people tend to move their body in the least energy expenditure. That is to say people will choose the straight line and when the walking speed is focused, people will choose the reasonable stride frequency to be little up and downs. What's more, the deviation of the liquid at straight direction decreases with the increasing speed. Since the background of the problem is a man who rushes out from office, we neglect the factor of the swing of hands which is opposite to the body because of the high walking speed, while when it comes to the robot, we take the swing of hands into account.

Since the possibility of spilling liquid interface is concerned with maximum acceleration, we choose it as the measurement of whether it will spill from the cup or not and calculate the maximum deviation angle at a given coffee level.

There are different types of coffee cup in our lives. So we select three types of cup particularly in this paper.



FIG: Three different types of cups. From left to right, the number is 1,2,3.

We are now faced with the followings:

- 1) Use the mathematic formula presents the process of human walking
- 2) Define the distribution of liquid under different force situation
- 3) When applied to the robot, the model should take the random vibration into account. The problem is how to measure the vibration.

Our approach is:

- 1) First, the process human walking is studied by theoretical analysis qualitatively. Then simplify the model and calculate the parameters by experiment.

- 2) Use the force balance model and simple hydromechanics equation to calculate the deviation angle of liquid.
- 3) According to different cups, we calculate the maximal capacity of coffee to test the possibility of the model under the focused stride length.
- 4) Through introducing the random vibration in y axis, we expand the model to three-dimensional space and apply it in the field of robot.

2. Models

2.1 Mathematically Modeling Human walking

2.1.1 Theoretical analysis

With no force or balance force, the trajectory of human walking is a uniform motion. When walking on the platform, the force in the vertical direction is in balance while in the horizontal direction, the force is friction force from one foot so that the motion is not a uniform motion. In conclusion, the trajectory of walking on the platform is alternate in accelerate motion and moderate motion.

Obviously, walking consists of steps so that the speed is a periodic function with the changing of time by a focused posture. Otherwise, because of the limitations of strength, the curve of human walking is a smooth one with no sudden change. To simplify the complicated biomechanics and physics process, we associate the procedure that the change of velocity by time varying with a simple harmonic motion which has average speed as the initial position.

2.1.2 Experiment

To simplify the process of human walking, we conduct an experiment to find the trajectory in the following steps:

- 1) Draw a two-meter-long ruler in the blackboard
- 2) Use a video camera to record the process of walking

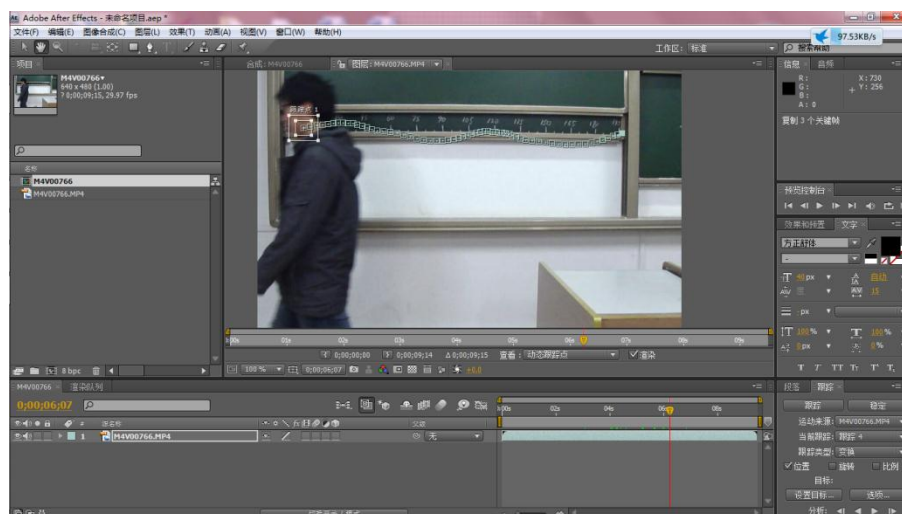


FIG 2-1 the process of experiment

- 3) Utilize dynamic track of Adobe After Effects to record the displacement per frame (30 frames per second) and calculate the velocity.
- 4) Create a scatter diagram based on the data of velocity and use the interpolation of Matlab
- 5) Generate the diagram

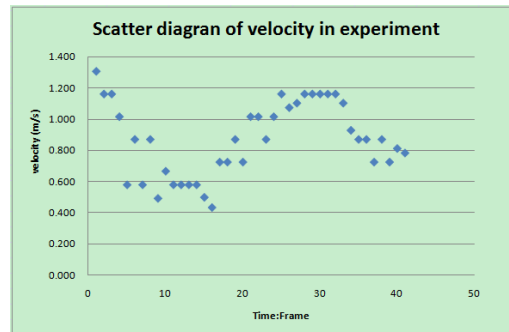


FIG 2-2 the scatter diagram of velocity in experiment

We find that the trajectory of human walking is similar to the simple harmonic motion so that we can simplify the process of human walking as simple harmonic motion.

2.1.3 Assumptions

- 1) The process of human walking is simplified as simple harmonic motion
- 2) There is no relative motion between the cup and body.
- 3) In the horizontal direction, the positive direction is the same as human walking while in the vertical direction, the positive direction is upward.
- 4) The liquid level maintains as a flat surface.

2.1.4 The model of human walking

● Description of human walking process

Based on the theory of simple harmonic motion and assumptions combined with common sense and reality, we decompose the human walking into two steps as the picture shows. In position1, the man with maximal horizontal velocity has the lowest height. In position 2, the man with the minimal horizontal velocity has the highest height. In conclusion, the process of human walking is a circle of these two steps.

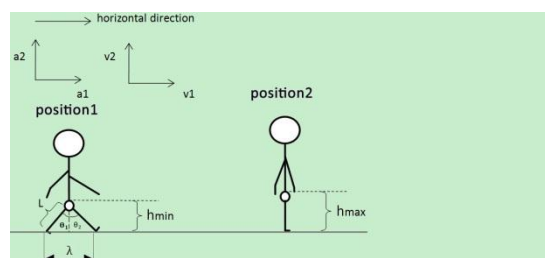


FIG 2-3 The period of human walking. In position1, the man has the maximal velocity and the lowest vertical height. In position2, the man has minimal velocity and the highest vertical height.

According to the analysis, we set the following parameters:

l : The length of leg

T : Period of human walking

ω : Angular velocity

θ : Intersection angle between leg and vertical direction

λ : Stride length

v_1 : Horizontal velocity

v_2 : Vertical velocity

a_1 : Horizontal acceleration

a_2 : Vertical acceleration

● Horizontal analysis

Suppose the angular velocity is ω and the period is T , then we have $\omega = \frac{2\pi}{T}$; average velocity is $\bar{v}$ and the step length is λ . The amplitude is k_1 which represents the harmony of human walking and is related to step length λ and compound factor p , then we have $k_1 = f(\lambda, p)$. The simple harmonic motion is[3]:

$$v_1(t) = \bar{v}_1 + k_1 \cos(\omega t + \varphi_1)$$

The accelerated speed a_1 equals to $\frac{dv_1(t)}{dt}$, we have

$$a_1 = -k_1 \omega \sin(\omega t + \varphi_1)$$

● Vertical analysis

Also, we decompose the human walking into two steps as the picture shows. When the velocity is maximum, the height is $h_{\min}$, otherwise the height is $h_{\max}$. Therefore, $h_{\max}$ is the length of leg and $h_{\min}$ is the length when the intersection angle of two legs is maximum[3].

$$h_{\max} = l$$

$$h_{\min} = \sqrt{l^2 - \left(\frac{\lambda}{2}\right)^2}$$

The vertical motion $h(t)$ is a simple harmonic motion. The center of vibration $b = \frac{1}{2}(h_{\min} + h_{\max})$ and the amplitude $k_2 = \frac{1}{2}(h_{\max} - h_{\min})$. The angular velocity is the

same as the horizontal direction while the phase position $\varphi_2 = \varphi_1$, then we have

$$h(t) = \frac{1}{2}(h_{\min} + h_{\max}) - k_2 \cos(\omega t + \varphi_2)$$

The accelerated speed in vertical direction is:

$$a_2 = \frac{d^2 h(t)}{dt^2} = k_2 \omega^2 \cos(\omega t + \varphi_2)$$

- **Synthetic analysis**

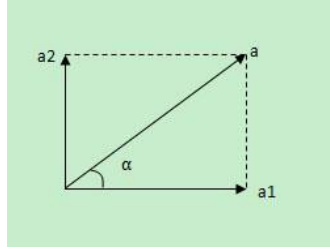


Fig 2-4 the sketch map of synthetic acceleration

To sum up, we obtain the acceleration of the rushing man[3].

$$a = \sqrt{a_1^2 + a_2^2}$$

$$\alpha = \arctan \frac{a_2}{a_1}$$

2.2 Mathematically Modeling Liquid Sloshing

If we decide to discuss the situation of change of the liquid in time varying when the man walks in detail, it probably involves the complicated Euler equation, Bernoulli differential equation and coefficient of kinematic viscosity.

According to the topic requirements, we could regard the coffee as a whole to simplify the problem. Then we could calculate the deviation angle through force analysis and some knowledge of hydromechanics.

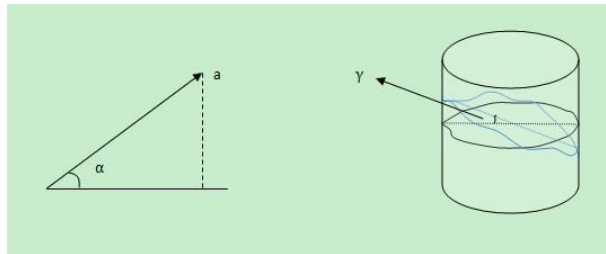


FIG 2-5 The sketch map of water sloshing.

On the basis of liquid equilibrium equation[2]:

$$dp = \rho(f_x dx + f_y dy + f_z dz)$$

Where p is air pressure, ρ is the density of water, f_x , f_y , f_z are three acceleration in different directions.

$$\begin{cases} f_x = -a \cos \alpha \\ f_y = 0 \\ f_z = -(g + a \sin \alpha) \end{cases}$$

Because the air pressure above the water is a constant, then we have:

$$dp = 0$$

$$-a \cos \alpha dx - (g + a \sin \alpha) dz = 0$$

$$-\frac{dz}{dx} = \tan \gamma = \frac{a \cos \alpha}{g + a \sin \alpha}$$

Where α is the intersection angel between acceleration and horizontal direction, which is given by the human walking model and γ is the deviation angle of the coffee.

2.3 Analysis of parameters and test of models

2.3.1. Analysis of ω

It's difficult to gain the deviation angle or the splash height when the man is walking with a cup, though we not necessarily evaluate directly. According to our formula, k_1 is the vibration width in physics while we guess this parameter is related to the stride length and the habit.

Theoretically, we can calculate the change of velocity under different stride length and frequency circumstance. However, k_1 is a complicated coefficient and if we change the value of k_1 , it is obvious the stride frequency will change so that it is not a test of single variable. Here, we only control the parameter ω to do some analysis and test.

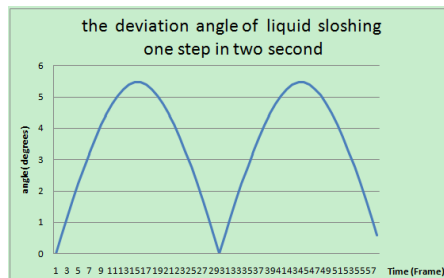


FIG 2-6(1)

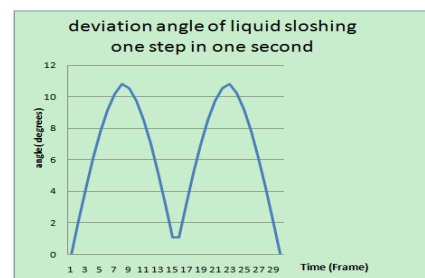


FIG 2-6(2)

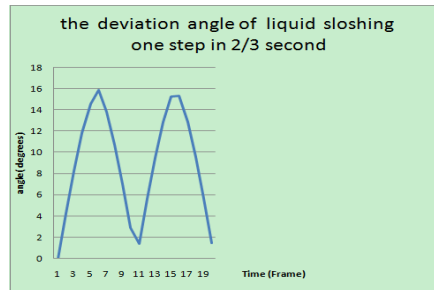


FIG 2-6(3)

FIG 2-6 We set three type of stride frequency: 1 step/s, 1step/2s, 1 step/0.67s, and calculate respectively using the model. We find that when the stride frequency increases (from (1)-(3)), the deviation angle of the liquid increases accordingly. When the stride frequency is minimal (1 step/2s), the maximum deviation angle is 5.48 degree. When the stride frequency is 1 step/s, the deviation angle increases to 10.81 degree. When the stride frequency is maximal (1 step/s), the deviation angle increases to 15.87 degree. Commonly, the consequence is in line with the reality. When people rush out (w increases), the splash height of coffee increases and the splash frequency speed up.

2.3.2. Test of acceleration

In order to identify the correctness of the model, we adopt the method of calculating the change of acceleration. We still use the three types of stride frequency.

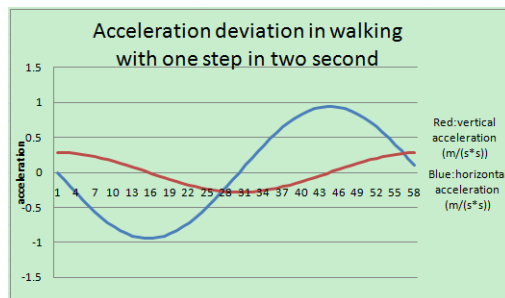


FIG 2-7(1)

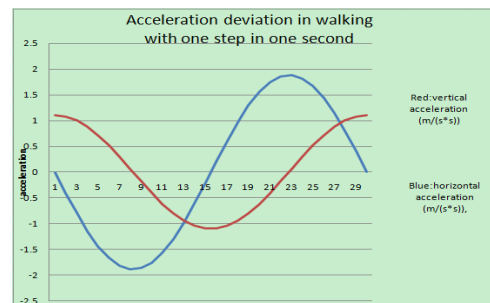


FIG 2-7(2)

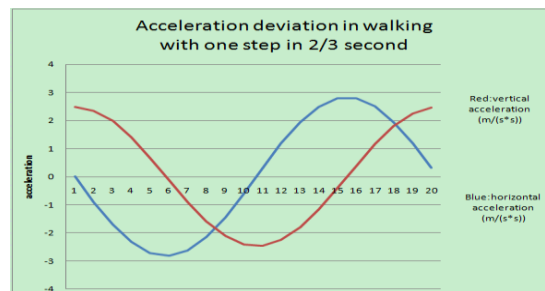


FIG 2-7(3)

FIG 2-7 We find that when the stride frequency increases, the range of horizontal acceleration increases through the contrast of the three blue lines in graphic because the longer the stride frequency, the faster the change of velocity. We also find that when the stride frequency increases, the range of vertical acceleration increases through the contrast of the three red lines in graphic. Because the displacement is a normal simple harmonic motion in the vertical direction and the period is the same as the horizontal direction, the stride frequency increases while the period in the

vertical direction decreases.

Therefore, our model is close to the real situation. It's reasonable to use the model to calculate the influence of stride length, stride frequency and other factors on the deviation angle of liquid.

2.4 Results

● Human walking

In terms of the parameter k_1 , because it is related to height, weight and other factors, it's hard to find the mathematic expression by theoretical analysis. We can calculate the value according to the focused man, length of step and period through experiment. On the basis of experiment, we get a group of values of velocity and time. For more accurate results, we use the cubic spline interpolation as follows:

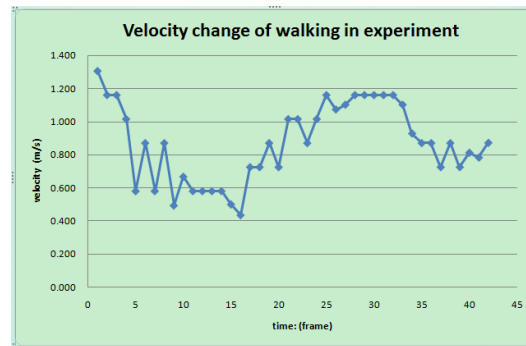


FIG 2-8 The scatter diagram. On the basis of our experiment, we note down the change of velocity in time varying. The experiment selects about 1.5 step length for 41 frames when the man is walking. We calculate the velocity through the situation per frame given by the Adobe after Effect. From the graphic, it is similar to the simple harmonic motion.

Then, in order to find the value of k_1 , we fit the curve by least square method as follows.(program is given in the appendix)

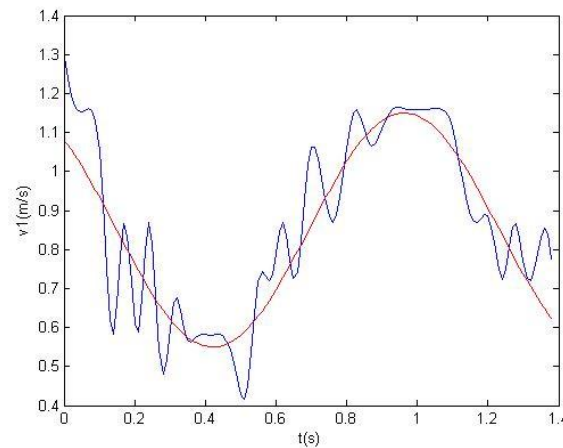


FIG 2-9 On the basis of the data from experiment, we obtain the chart by the interpolation

and fitting of the curve and calculate the parameters.

Based on a series data of a normal university student (1.75-meter-height, 70-kilogram-weight), we calculate the compound parameter $k_1=0.30$, the average velocity $\bar{v}=0.85$, the angular velocity $\omega=5.80$, the initial position $\varphi_1 = 0.70$.

In horizontal:

$$v(t) = 0.85 + 0.30 \cos(5.80t + 0.70)$$

$$a_1 = -1.74 \sin(5.8t + 0.7)$$

In vertical:

$$h(t) = 0.97 - 0.030 \cos(5.80t + 0.7)$$

$$a_2 = \cos(5.80t + 0.7)$$

In synthetic:

$$a = \sqrt{3 \sin^2(5.8t + 0.7) + \cos^2(5.80t + 0.7)}$$

$$\alpha = \arctan \frac{\cos(5.80t + 0.7)}{-1.74 \sin(5.8t + 0.7)}$$

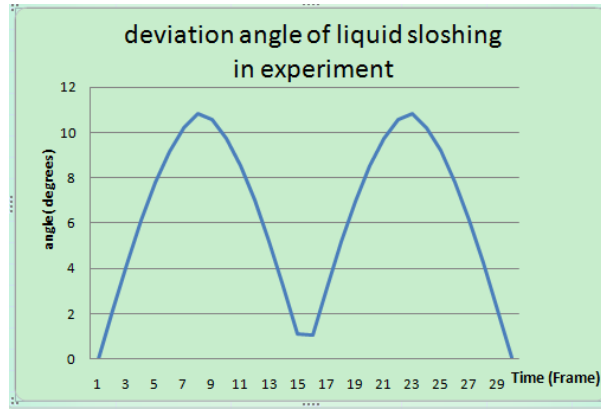
● Water sloshing

Based on experimental data, we can obtain the unknown parameters a and α . Using the theoretical formula of liquid sloshing we inferred before, we can have the deviation angle:

$$\gamma = \arctan \frac{a \cos \alpha}{g + a \sin \alpha}$$

$$\Rightarrow \begin{cases} \gamma = \arctan \frac{a \cos \alpha}{g + a \sin \alpha} \\ a = \sqrt{3 \sin^2(5.8t + 0.7) + \cos^2(5.80t + 0.7)} \\ \alpha = \arctan \frac{\cos(5.80t + 0.7)}{-1.74 \sin(5.8t + 0.7)} \end{cases}$$

Also, we can gain the graphic of the deviation angle in time varying as follows:



Maximum: $\gamma = 10.81^\circ$

3. Conclusions

In terms of the given diameter of the cup, we can get the critical value of splash height in the following chart:

Type of cup	Diameter (m)	Splash height (m)	Capacity
1	0.07	0.12	94.82%
2	0.055	0.18	97.28%
3	0.08	0.09	92.10%

4. Applications in robots

4.1. Mathematically Modeling Liquid Sloshing

4.1.1. The explanation of frame of axes

x: the direction of robot walking

y: the direction which obey to the right hand ruler

z: upwards

4.1.2. Analysis in y axis

When it comes to robot, we expand the model, making it to more valuable for the differences between human beings and robots. Expect for the horizontal and vertical simple harmonic motion, the random vibration in y axis should be taken into account. The vibration in y axis is related to stride length and frequency. Usually, the higher the stride frequency, the lower the probability of vibration in y axis will occur. What's

more, the shorter the stride length, the smaller the vibration in y axis will be.

To handle the vibration in y axis, we simplify the process as linearity, the format as follows:

$$a = m \times \frac{1}{\omega} + n \times \frac{\lambda}{2}$$

4.1.3. Analysis in x axis

In the previous model, we simplify the model of human walking as a simple harmonic. However, the amplitude is given by the experiment and fitting curve which is related to the stride length and habit. It's necessary to consider the influences of different stride frequency and length on the parameter k1. Though we account K as a complicated fluctuant parameter, we don't rule out the possibility that K is a constant. In order to discuss the value of K, we set a focused k1 and a changing stride length and frequency to calculate the varying of acceleration in unit stride length.

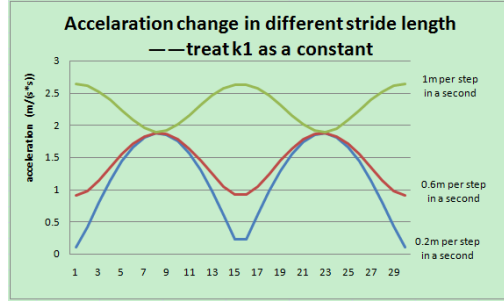


FIG 4-1 The situation of change under the focused circumstance. When the k1 and stride frequency is focused, we select three types of stride length: 0.2/m, 0.6/m, 1.0/m and calculate the change of acceleration. We find that it's hard to account for the curve when the stride length is 1.0/m.

Through the experiment, we have a suspicion whether k1 is related to the stride length. Although the increasing of stride length influences the simple harmonic motion in z axis, the vibration in z axis is slight. If the change of horizontal acceleration has nothing to do with the stride length, when the stride length is longer and the frequency is the same, the vibration of coffee is almost the same. It is not in line with the common sense.

Therefore, we suppose the parameter k1 is related to the stride length and test the supposal. We suppose that k1 is the half of stride length and the consequence is as follows:

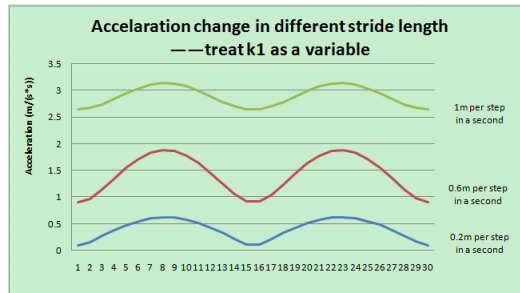


FIG 4-2 the acceleration increases when the stride length increases by a focused stride frequency. Although we are not sure the specific quantitative relation between stride length and k1, they are definitely positive correlations.

4.1.4. Analysis in general

When it comes to the influence of accelerations on the deviation of coffee, the deviation angel is difficult to distinct. However, we just need to concern whether the coffee will spill instead of the direction of acceleration.

Because the possibility of the vibration in y axis is random, we add the vibration to the every time point during a period to calculate the safe distance between the cup interface and the maximal deviation height.

Defining the stride length and frequency of the robot, through processing in the different assemble of stride length and frequency, we calculate the average velocity, the maximal capacity of coffee and the maximal deviation angel.

4.2. Results

During the process of robot walking under the focused leg length (1m), the results of velocity, maximal deviation angle and maximal capacity of coffee as follows:

time per step(s)	length per step(m)	average velocity (m/s)	maximum angel of coffee (degrees)	space usage: height& volume(%)	space usage: height& volume(%)	space usage: height& volume(%)
2	0.4	0.2	4.75	97.59%	98.74%	96.32%
1	0.4	0.4	7.57	96.16%	97.99%	94.15%
0.67	0.4	0.6	10.96	94.46%	97.10%	91.55%
2	0.6	0.3	6.55	96.68%	98.26%	94.93%
1	0.6	0.6	11.13	94.37%	97.05%	91.42%
0.67	0.6	0.9	16.06	91.94%	95.78%	87.71%
2	0.8	0.4	8.38	95.75%	97.78%	93.53%
1	0.8	0.8	14.58	92.66%	96.16%	88.82%
0.67	0.8	1.2	20.80	89.65%	94.58%	84.22%
2	1	0.5	10.20	94.84%	97.30%	92.13%
1	1	1	17.88	91.05%	95.31%	86.36%
0.67	1	1.5	25.13	87.62%	93.52%	81.14%
2	1.2	0.6	12.00	93.94%	96.82%	90.76%
1	1.2	1.2	21.00	89.55%	94.53%	84.08%
0.67	1.2	1.8	28.99	85.87%	92.60%	78.47%

When the stride length increases, the change of capacity of cup is as the following chart:

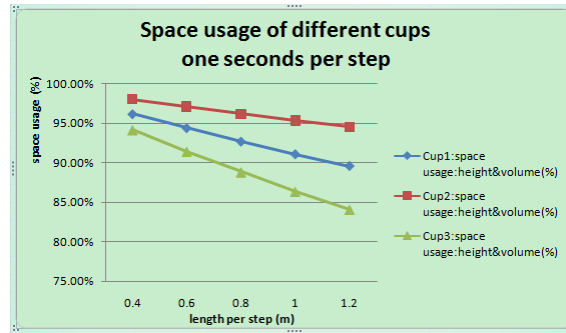


FIG 4-3 from the chart, we find that the capacity of the third cup is small and decreases rapidly with the increasing of stride length. The change of the first and second cup is small in general. The reason is obvious. The height of the cup is shorter while the diameter is longer. When the deviation angle is focused, the safe height will be set higher. Meanwhile, the height of the third is short and the ability of holding coffee is weaker.

The first and second cups belong to the type of slender and the second cup is more slender than the first cup. Through the chart, we find that the more slender the cup, the stronger the ability of holding coffee. That is to say, when we are walking in a rush, we don't need to worry the spilling of coffee.

Reconsidering the robot in reality, we select two types of cup: the slender one and dumpy one from the chart to generate the suitable walking strategy. Under the suitable strategy, the robots don't worry the spilling of coffee in its maximal velocity and guarantee the surplus of coffee is over 90%

type of cup	time per step (s)	length per step (m)	average velocity (m/s)	maximum angel of coffee (degrees)	Cup1: space usage: height& volume (%)
Long thin cup (e.g. $r=0.028\text{m}$ $h=0.18\text{m}$)	0.67	1.2	1.8	28.99	92.60%
short wide cup (e.g. $r=0.04\text{m}$ $h=0.09\text{m}$)	1	0.6	0.6	11.13	91.42%

For the slender one, we design a fierce walking strategy: stride length is 1.2m, 0.67s per step, the capacity of coffee is 92.6%. The maximal deviation angle is 28.99 degree. Of course, don't worry, under the calculation, the coffee will not spill.

For the dumpy one, we design a conservative walking strategy: stride length is 0.6m, 1s per step, the capacity of coffee is 91.42%. The maximal deviation angle is 11.13 degree. The whole process is smoother opposite to the slender one.

5. References

- [1]Ronan Boulic, A global human walking model with real-time kinematic personification. 1990, Volume 6, Issue 6, pp 344-358
- [2]Bruce R.Munson, Fundamentals of Fluid Mechanics. 7-121-03327-5
- [3]A.P.French, Newtonian mechanics. /0.55(1),0.88(2),0.77(3)

6. Appendix

6.1. program: fitting parameters

```
clear all
min=[0.5, 0, 5.0, -1];
max=[1.5, 0.35, 7.5, 1];
best=min;
d=inf;
p=[];
x=[1:41];
x=(x-1).*(1/29);
y(1,:)= [1.305000000000000,1.160000000000000,1.160000000000000,1.015000000000000,0.5800000
00000000,0.870000000000000,0.580000000000000,0.870000000000000,0.493000000000000,0.6
670000000000000,0.580000000000000,0.580000000000000,0.580000000000000,0.5800000000000
000,0.500000000000000,0.435000000000000,0.725000000000000,0.725000000000000,0.870000
000000000,0.725000000000000,1.015000000000000,1.015000000000000,0.870000000000000,1.01
500000000000000,1.160000000000000,1.073000000000000,1.102000000000000,1.160000000000000,1.1
600000000000000,1.160000000000000,1.160000000000000,1.160000000000000,1.102000000000000,0.
928000000000000,0.870000000000000,0.870000000000000,0.725000000000000,0.870000000000
000,0.725000000000000,0.812000000000000,0.783000000000000];
xi=[0:0.01:1.38];
y=interp1(x,y(1,:),xi,'spline');
x=xi;
[aa a]=size(x);
for i=(2:a-1)
    if (y(i)<y(i-1)&& y(i)<y(i+1))||(y(i)>y(i-1)&& y(i)>y(i+1))
        p=[p i];
    end
end
[aa a]=size(p);
temp=0;
%for i=2:a-1
% [zz z]=size(xi);
```

```

%     ii=temp+i;
%     xi=[xi(1:ii-1) (xi(ii)-0.0001):0.00001:(xi(ii)+0.0001) xi(ii+1:z)];
%     temp=temp+20;
%end

%yi=interp1(x,y(1,:),xi,'spline');
x=xi;
%y=yi;
for a1=min(1):0.05:max(1)
    for a2=min(2):0.01:max(2)
        for a3=min(3):0.1:max(3)
            for a4=min(4):0.1:max(4)
                c=[a1 a2 a3 a4];
                y(2,:)=c(1)+c(2)*cos(c(3)*x+c(4));
                dd=sum(abs(y(1,:)-y(2,:)));
                if dd<d
                    d=dd
                    best=c
                    zz=y(2,:);
                end
            end
        end
    end
end
end
plot(x,y(1,:), 'b')
hold on
plot(x,zz, 'r')
xlabel('t(s)')
ylabel('v1(m/s)')
%hold on
%z=(0:0.01:1);
%zz=best(1)+best(2)*sin(best(3)*z+best(4));
%plot(z,zz,'g')

```